Neural Operator Surrogate for Space Charge in Particle Accelerator Simulations

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***Abstract*—**

I. Introduction

CHEETAH is a fully-differentiable beam dynamics code [1] whose three-dimensional space charge kick [2] is costly in computation time and in terms of the memory needed for reverse-mode differentiation [3]. Most of this cost arises in the grid-based field solve, which maps the charge density of the beam to its electrostatic potential. This work investigates whether a Fourier neural operator (FNO) [4] can replace this field solve such that the computation time and the memory of the forward and backward pass decrease, while the relative error of the potential stays below WERT % and the beam moments deviate by less than WERT % from the reference solver.

A. Cheetah and space charge

CHEETAH is written in PyTorch and built as a modular framework with lattice elements that map an incoming to an outgoing beam, so that arbitrary beamlines are assembled by composing these elements. It is presented in the paper [1] alongside use-cases regarding bayesian optimization, reinforcement learning, system identification, implementation of a neural netowrk surrogate and gradient-based beam tuning and optimization. What additionally makes CHEETAH distinct, is that it shortens simulations by up to three orders of magnitude compared to other Frameworks like ASTRA, OCELOT and XSUITE. [1, S. 6] Since all operations are PyTorch operations, gradients of beam properties with respect to lattice and beam parameters follow from automatic differentiation, which enables gradient-based tuning and system identification [1]. Dhamrait et al. [3] added a three-dimensional space charge kick based on the particle-in-cell (PIC) method. Training a surrogate for this kick is the subject of this work.

Space charge denotes the Coulomb interaction among the particles of a beam. Its self-field defocuses the beam, with the magnetic contribution partially cancelling the electric one by a factor $1/\gamma^2$, where γ is the Lorentz factor [2]. It is therefore most pronounced for high bunch charge at low energy and drives, e.g., emittance growth and intensity-dependent beam loss [3]. Because the force depends on the distribution of the particles within the beam, space charge cannot be written as a transfer map like other elements in CHEETAH. It is applied instead as a sequence of kicks: between two kicks the beam is transported by the element map, and each kick changes the momentum of every macroparticle by the self-field force integrated over the time Δt in which the beam travels the step Δs [3]. A direct summation over all particle pairs scales as $\mathcal{O}(N^2)$ for N macroparticles. PIC methods instead deposit the charge on a grid, solve for the field on the grid, and interpolate the force back to the particles [5], [6]. A kick thus consists of a particle part, whose cost grows with N, and a grid part, whose cost grows with the number of cells N_{cells} [3].

Table I lists the computation time of one space charge kick in CHEETAH, for each functional step. For 10^4 macroparticles on 32^3 cells, the forward pass takes $46.0\,\text{ms}$ and the backward pass $41.5\,\text{ms}$. Of the forward time, $28.7\,\text{ms}$ ($62\,\%$) are spent evaluating the integrated Green function (IGF) and $3.2\,\text{ms}$ ($7\,\%$) on the FFT-based convolution; on 64^3 cells the IGF share rises to $78\,\%$. The IGF depends on the cell size and γ. Charge deposition and force interpolation, in contrast, grow with N and account for $60\,\%$ of the forward time at 10^5 macroparticles on 32^3 cells. For comparison, a linear element tracks the same beam in WERT ms. For the backward pass, PyTorch stores intermediate results of every kick until the gradient is computed. Dhamrait et al. [3] measured that this memory grows linearly with N and N_{cells} and proportionally to the number of kicks, with $1.4\,\text{kB}$ per macroparticle, dominated by deposition and interpolation, and $1.2\,\text{kB}$ per cell, dominated by the IGF evaluation. For 10^4 macroparticles on 32^3 cells this amounts to about $53\,\text{MB}$ per kick, i.e. WERT GB for a beamline with WERT kicks. Because time and memory of the computations on the grid grow with N_{cells}, the grid resolution trades cost against the accuracy of the field ZITAT – Beleg fuer: Genauigkeit vs. Gitteraufloesung; oder eigene Aufloesungsstudie, Abschnitt ABSCHNITT.

B. Fourier neural operators

Neural operators learn mappings between function spaces instead of between finite-dimensional vector spaces [4], [7], [8] compared to conventional neural networks. In the present case, the input function is the charge density ρ and the output function the potential ϕ. The solution operator $\mathcal{G}\colon \rho \mapsto \phi$ is linear and given by the convolution with the Green function (Eq. (3)). A conventional network, e.g. a convolutional network, approximates this map for one fixed grid, and its error grows when the resolution changes [4]. A neural operator parametrises $\mathcal{G}$ itself, so that its parameters do not depend on

TABLE I

FORWARD AND BACKWARD COMPUTATION TIME OF ONE SPACE CHARGE KICK IN CHEETAH [1], [3], IN MS, BROKEN DOWN BY STEP (FORWARD PASS). CPU WERT: MODELL, 4 KERNE, FLOAT32, CHEETAH WERT, PYTORCH WERT. IGF: EVALUATION OF THE INTEGRATED GREEN FUNCTION ON THE DOUBLED GRID; FFT: CONVOLUTION; DEP.+INT.: CHARGE DEPOSITION AND FORCE INTERPOLATION. WERT: MEDIAN UEBER N WIEDERHOLUNGEN, STREUUNG.

Grid	N	IGF	FFT	Dep.+Int.	Fwd	Bwd
32^3	10^4	28.7	3.2	8.6	46.0	41.5
32^3	10^5	20.7	1.7	57.7	95.8	120.3
64^3	10^4	136.8	22.7	8.5	174.4	134.4
64^3	10^5	116.0	22.7	85.0	240.8	216.2

the discretisation. Thus one trained model can be evaluated on different grids, provided the grid resolves the relevant features [4], [7]. The FNO is a special form of Neural Operator. Its working principle is well-explained in the work of Li et al. at [4, p.ȧ ff., especially Figure 2]. The FNO lifts the input map $a(x)$ (here ρ) to d_v channels via Neural Network P, applies T Fourier layers

$$v_{t+1}(\mathbf{x}) = \sigma\big(W v_t(\mathbf{x}) + \mathcal{F}^{-1}\big(R \cdot \mathcal{F}(v_t)\big)(\mathbf{x})\big)\,, \qquad (1)$$

and projects the result to the output map $u(x)$ (here ϕ). Here $\mathcal{F}$ is the Fourier transform, R a learned complex weight tensor acting on the lowest $k_{\max}$ modes per dimension, W a learned pointwise linear map, and σ a nonlinear activation. With FFTs, one layer costs $\mathcal{O}(n \log n)$ for n grid points [4]. Li et al. [4] report consistent errors across resolutions and evaluation on finer grids than used in training (zero-shot super-resolution).

The FNO fits the space charge problem because it shares the structure of the classical solver. CHEETAH computes the potential as in Eq. (5) on a doubled grid, while a Fourier layer computes $\mathcal{F}^{-1}\big(R \cdot \mathcal{F}(v)\big)$. A single Fourier layer on the doubled grid, without nonlinearity, local term and mode truncation and with $R = \mathcal{F}(\tilde{G})$, thus reproduces the solver exactly. Thus the FNO is trained to replace the evaluation of the IGF (Eq. (4), which dominates time and memory of the kick for moderate number of particles N (Sec. I-A). Additionally it is discretisation invariant, i.e. it allows a refinement of the *grid* without retraining and without affecting the computational parts of the kick, whose costs grow with N. Li et al. also attribute to the local term W the ability to represent non-periodic boundary conditions [4], shown for problems with bounded domains. Whether this extends to the open boundaries of the space charge problem, so that the grid doubling of the classical solver can be omitted, is tested in this work (Sec. II-A).

II. METHODOLOGY

A. Architecture of the FNO surrogate within Cheetah

A space charge kick over the step Δs consists of the following steps [3]:

1) The macroparticles are transformed from coordinates at fixed s to coordinates at fixed time t.
2) Their charge is deposited on a Cartesian grid of 32^3 cells spanning $\pm 3\sigma$ of the beam in each dimension by trilinear (cloud-in-cell) weighting [5], which yields the charge density ρ_{ijk}.
3) The potential follows from the modified Poisson equation [9]

$$\partial_x^2\phi + \partial_y^2\phi + \frac{1}{\gamma^2}\partial_z^2\phi = -\frac{\rho}{\epsilon_0}, \qquad (2)$$

solved in the laboratory frame by convolution with the IGF [3], [10]

$$\phi_{ijk} = \sum_{i'j'k'} \rho_{i'j'k'}\, G(\mathbf{x}_{ijk} - \mathbf{x}_{i'j'k'}). \qquad (3)$$

G is the Green function of Eq. (2) integrated over one cell. It is evaluated in closed form from an antiderivative g at the eight corners of the cell [11],

$$G(\mathbf{x}) = \sum_{a,b,c\in\{\pm1\}} abc\; g\Big(x + a\tfrac{\Delta x}{2},\, y + b\tfrac{\Delta y}{2},\, \gamma z + c\tfrac{\gamma\Delta z}{2}\Big)\,. \qquad (4)$$

Since Eq. (3) costs $\mathcal{O}(N_{\text{cells}}^2)$ when evaluated directly, CHEETAH uses the Hockney method [6]: ρ is zero-padded and G is evaluated on a grid doubled in every dimension ($\tilde{\rho}$, $\tilde{G}$, 64^3 points), so that the cyclic convolution computed by the FFT equals the linear convolution of Eq. (3), i.e. open boundaries:

$$\phi = \mathcal{F}^{-1}\big(\mathcal{F}(\tilde{G}) \cdot \mathcal{F}(\tilde{\rho})\big)\Big|_{32^3}. \qquad (5)$$

4) The force $\mathbf{F} = -q\nabla\phi/\gamma^2$ is computed on the grid by central differences [3].
5) $\mathbf{F}$ is interpolated trilinearly to the macroparticles.
6) Each momentum is updated, $\mathbf{p} \leftarrow \mathbf{p} + \mathbf{F}\Delta t$ with $\Delta t = \Delta s/(\beta c)$, and the coordinates are transformed back to fixed s.

The surrogate replaces the evaluation of $\tilde{G}$ and the convolution of Eq. (5) within step 3, and maps ρ on 32^3 cells to ϕ on the same grid. Steps 1, 2 and 4–6 remain unchanged, and so does the interface of the kick element. the surrogate kick can therefore be placed in any CHEETAH beamline. The replaced routine is given in Listing 1 in the appendix.

B. Computation Time and Memory Usage of Space Charge

III. RESULTS

IV. CONCLUSION

APPENDIX

A. Code Snippets of FNO in Cheetah

Listing: Poisson-Loesung mit FNO statt IGF-Faltung

Fig. 1. Replaced field solve of the space charge kick.

BILD: Zeit je Kick ueber Zellzahl, alle Konfigurationen

Fig. 2. Bildunterschrift

B. Computation time and memory measurement plots

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